Northwestern

# Wildcat Beamer Theme

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# Introduction

The Wildcat theme is a Beamer theme for Northwestern University that can be easily customized with different colors, fonts, and background patterns.

It is inspired by the Metropolis theme by Matthias Vogelgesang, and incorporates the Northwestern University facet design pattern. Licensed under the GNU GENERAL PUBLIC LICENSE.

# Contents

Background Patterns

Colors

Fonts

Block Environments

Special Frames
    Subsection Title

Advanced Customization

# 1

# Background Patterns

# Background Patterns

The theme ships with 12 built-in background patterns. The easiest way to switch is via a theme option:

```
\usetheme[pattern=triangle]{wildcat}
```

Section pages automatically use the same pattern as the title page. Available pattern options:

- `facet` — NU facets (default)
- `facetrand` — NU facets (alternate)
- `triangle` — triangle mosaic
- `triangleiso` — isometric triangles
- `trianglemesh` — wireframe mesh
- `squares` — square grid
- `circles` — circles
- `hexraise` — raised hex clusters
- `hexrand` — random hex shading
- `network` — network grid
- `quilt` — quilt rings
- `plain` — solid color

See `wildcat-gallery.pdf` for a visual preview of all patterns.

# 2

# Colors

# Colors

The theme has a few Northwestern-specific colors defined, which you can use in your slides. These are:

- nupurple
- nupurple90
- nupurple80
- nupurple70
- nupurple60
- nupurple50
- nupurple40
- nupurple30
- nupurple20
- nupurple10
- nupurple160
- nupurple150
- nupurple140
- nupurple130
- nupurple120
- nupurple110
- nurichblack
- nubrightgreen
- nubrightteal
- nubrightblue
- nubrightyellow
- nubrightorange
- nubrightred
- nudarkgreen
- nudarkteal
- nudarkblue
- nudarkyellow
- nudarkorange
- nudarkred

# Modifying the Primary Color

Pass a `color` option to `\usetheme` — the theme auto-generates all lighter and darker shades automatically:

```
\usetheme[color=nudarkblue]{wildcat}
```

You can combine `color` and `pattern` in a single line:

```
\usetheme[color=nudarkblue, pattern=hexraise]{wildcat}
```

The `nu*` colors from the previous slide are all valid choices, as are the standard xcolor built-ins (`red`, `blue`, `cyan`, etc.). The alert and example colors (`wcalerted`, `wcexample`) are not affected and can be overridden separately.

# Modifying the Primary Color: Custom RGB

The `color` option accepts any *named* color — not raw RGB values. If your brand color isn't already defined, declare it first:

```
\definecolor{myblue}{RGB}{0,53,107}
\usetheme[color=myblue]{wildcat}
```

When you pass a color via the `color` option, that color becomes available to use as `wcprimary` in your slides, and the theme auto-generates the lighter and darker shades as `wcprimary10`–`40` and `wcprimary110`–`140`, respectively. These are also available as named colors you can use in your slides, and the theme uses them internally for block backgrounds and pattern fills.

The auto-generated shades follow a simple scale: `wcprimary10`–`40` mix the base color with white (lighter tints), and `wcprimary110`–`140` mix it with black (darker shades). These are sufficient for most use cases — if you need precise per-shade control, see the Advanced Customization section at the end of this document.

# Using Colors in Graphs

## Python

```
import matplotlib.pyplot as plt
import pandas as pd
plt.style.use('ggplot')
# Create colors
wcprimary = (78/255,42/255,132/255)
df = pd.read_stata("source/graphs/auto.dta")
plt.scatter(df["weight"], df["mpg"], color=wcprimary)
plt.xlabel("Weight")
plt.ylabel("MPG")
plt.savefig("source/graphs/plot-python.pdf")
plt.show()
```

# Using Colors in Graphs

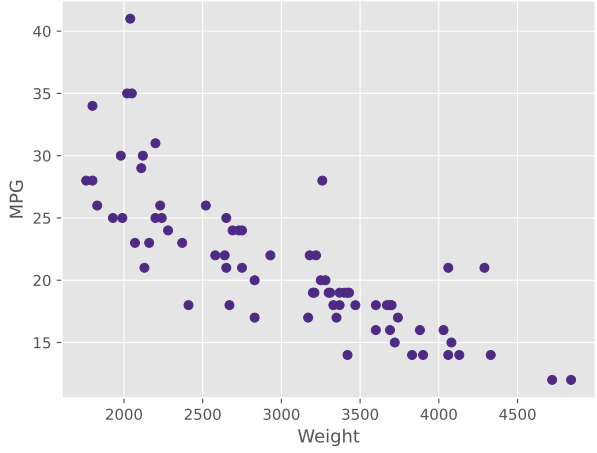



Figure: Python Graph Example

# Using Colors in Graphs

R

```
library(ggplot2)
library(haven)
# Load data
df <- read_dta("source/graphs/auto.dta")
# Define custom RGB colors
wcprimary <- rgb(78/255, 42/255, 132/255)

# Plot mpg against weight, with marker color wcprimary
plot <- ggplot(df, aes(x = weight, y = mpg)) +
    geom_point(color = wcprimary) +
    labs(x = "Weight", y = "MPG")

# Save the plot to a file with a specific size
ggsave("source/graphs/plot-R.pdf", plot, width = 8, height = 5)
```

# Using Colors in Graphs

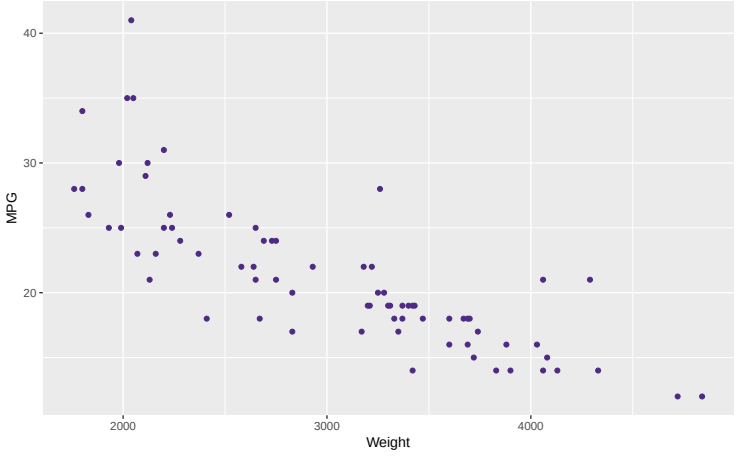



Figure: R Graph Example

# Using Colors in Graphs

## Stata

```
ssc install schemepack, replace
sysuse auto, clear
local wcprimary "78 42 132"

twoway (scatter mpg weight, mcolor("`wcprimary'")), ///
        scheme(gg_tableau)

graph export "source/graphs/plot-stata.pdf", replace
```

# Using Colors in Graphs

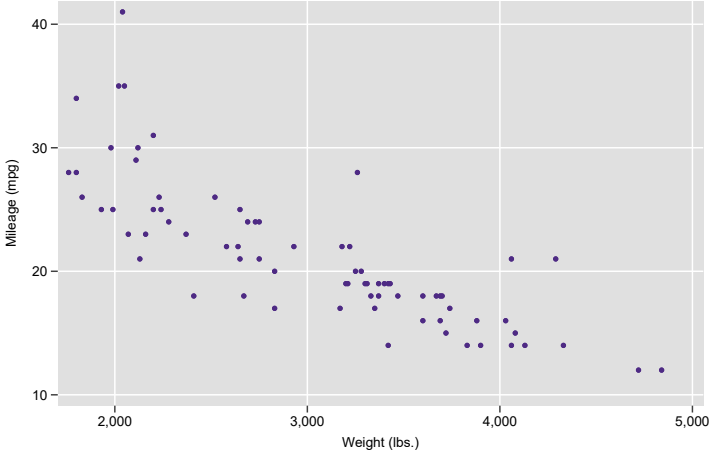



Figure: Stata Graph Example

# 3

# Fonts

# Fonts: Default (Fira Sans)

By default, the theme uses **Fira Sans** — a clean, geometric sans-serif that ships with TeX Live and MiKTeX and is pre-installed on Overleaf. No font files need to be downloaded or bundled:

```
\usetheme{wildcat}          % Fira Sans loads automatically
```

**Overleaf:** works out of the box.
**MiKTeX:** auto-installs on first compile.
**TeX Live:** if the font is not found, run `tlmgr install fira` in your terminal, then recompile.

Requires **XeLaTeX** or **LuaLaTeX** — PDFLaTeX will fail.

# Fonts: Poppins + IBM Plex Sans

For a closer match to Northwestern's recommended free fonts, use `font=poppins`. Download both families from Google Fonts and place them in a `fonts/` folder next to your `.tex` file:

```
\usetheme[font=poppins]{wildcat}
```

```
your-presentation/
  your-slides.tex
  fonts/
    Poppins/
      Poppins-Medium.ttf     Poppins-MediumItalic.ttf
      Poppins-SemiBold.ttf   Poppins-SemiBoldItalic.ttf
      Poppins-Light.ttf      Poppins-LightItalic.ttf
      Poppins-Regular.ttf    Poppins-Italic.ttf
    IBM_Plex_Sans/
      IBMPlexSans-Light.ttf       IBMPlexSans-LightItalic.ttf
      IBMPlexSans-Regular.ttf     IBMPlexSans-Italic.ttf
```

For proprietary Northwestern brand fonts (Campton + Akkurat Pro), see the Advanced Customization section.

# Font Style: Examples

Here is what the body font will look like under normal usage:

- Regular
- *Italic*
- **Bold**
- ***Bold Italic***
- Alert
- *Alert Italic*
- Math:

$$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n$$

# 4

# Block Environments

# Pattern Blocks (tcolorbox)

Three patterned block environments are available as drop-in replacements for standard Beamer blocks: `tblock`, `talert`, and `texample`.

**T Block Title**

This is a tcolorbox style block.

**T Alert Title**

This is a tcolorbox style alert block.

**T Example Title**

This is a tcolorbox style example block.

# Custom Color Pattern Blocks

`tpatternbox` is a variant that accepts an optional color argument, letting you use any named color for the block header.

**Custom Pattern Block Title**

This is a tpatternbox block. It can be created with the following code:

```
\begin{tpatternbox}[nudarkyellow]{Custom Pattern Block Title}
    This is a tpatternbox block...
\end{tpatternbox}
```

# Box Examples (Default)

Standard Beamer blocks work as usual. The theme styles them with flat (non-rounded, non-shaded) colors.

**Main Block**

This is an example block

**Alert Box**

This is an alert box

**Example Box**

This is an example box

# 5

# Special Frames

# Sections

`\section{}` groups slides and automatically inserts a section title slide. Pass `sectionpage=none` to `\usetheme` to disable them:

```
\usetheme[sectionpage=none]{wildcat}
```

# Subsections

The normal `\subsection{}` command creates subsections, and there is an option to create a subsection title slide (disabled by default). To enable them, use the `subsectionpage=on` option when loading the theme:

```
\usetheme[subsectionpage=on]{wildcat}
```

# 5.1

# Special Frames

*Subsection Title*

The `[plain]` option produces a fully blank frame with no header, footer, or background. Use it for title cards, full-bleed figures, or transition slides:

```
\begin{frame}[plain]{}
...
\end{frame}
```

# Standout Slides

The `[standout]` frame option creates a full-pattern slide with centered white text — useful for section breaks, closing slides, or emphasis. No manual formatting needed:

```
\begin{frame}[standout]
  Appendix
\end{frame}
```

The legacy `\standout{text}` command is still supported for backward compatibility.

# Appendix

# 6

# Advanced Customization

# Advanced: NU Brand Fonts (Bundled)

Northwestern affiliates can use the official brand fonts — **Campton** (headings) and **Akkurat Pro** (body). These are proprietary; obtain them from `northwestern.edu/brand`. Bundle the OTF files in a `fonts/` folder and use:

```
\usetheme[font=nu]{wildcat}
```

```
your-presentation/
  your-slides.tex
  fonts/
    Campton/
      Campton Medium.otf    Campton Light.otf
    Akkurat/
      AkkuratProLight-Regular.otf  AkkuratProLight-Italic.otf
      AkkuratPro-Regular.otf       AkkuratPro-Bold.otf
```

# Advanced: NU Brand Fonts (System-Installed)

If Campton and Akkurat Pro are already installed as system fonts (visible in Word, etc.), no `fonts/` folder is needed. Use:

```
\usetheme[font=installed]{wildcat}
```

fontspec loads the fonts directly from the OS font registry — no `fonts/` folder needed. This is convenient for local compilation, but won't work on Overleaf or when sharing with collaborators who don't have the fonts installed. In those cases, use `font=nu` with bundled OTF files as described on the previous slide.

# Advanced: Fine-Grained Color Control

The auto-generated shades are a good starting point, but for precise brand colors you can override individual shades manually. Redefine any of the `wcprimary*` colors *after* `\usetheme`:

```
\usetheme{wildcat}                          % or with color= option
\definecolor{wcprimary}{RGB}{0,53,107}      % Yale Blue
\definecolor{wcprimary140}{RGB}{0, 34, 70}
\definecolor{wcprimary130}{RGB}{0, 40, 80}
\definecolor{wcprimary120}{RGB}{0, 45, 91}
\definecolor{wcprimary110}{RGB}{0, 50, 102}
\definecolor{wcprimary40}{RGB}{153, 174, 196}
\definecolor{wcprimary30}{RGB}{179, 194, 211}
\definecolor{wcprimary20}{RGB}{204, 215, 225}
\definecolor{wcprimary10}{RGB}{230, 235, 240}
```

Only the shades actually used by the theme need redefining: `wcprimary10`–`40` (block bodies) and `wcprimary110`–`140` (facet pattern segments).

# Advanced: Custom Background Pattern (I)

To create your own pattern, define a command that draws TikZ commands filling the slide. The command should take one argument — the color — so that the theme can reuse it in different contexts. Here is a minimal example:

```
\newcommand{\mypattern}[1]{
    \fill[color=#1] (0,0) rectangle (\paperwidth,\paperheight);
}
```

The pattern command is called inside a `tikzpicture` environment whose bounding box already covers the full slide, so you only need to provide the drawing commands — no `\begin{tikzpicture}` wrapper is needed.

Any valid TikZ commands work. The built-in patterns each take one color argument and use `\paperwidth` / `\paperheight` to scale to the slide dimensions automatically.

# Advanced: Custom Background Pattern (II)

Once your pattern command is defined, register it with the theme using two lines in your preamble *after* `\usetheme{wildcat}`:

```
\renewcommand{\bgpattern}{\mypattern{wcprimary}}
\renewcommand{\bgpatternblock}[1]{\mypattern{#1}}
```

Two commands are needed because the theme uses the pattern in two different ways:

- `\bgpattern` — title page, section slides, and standout frames. The color is fixed, so you bake it in when registering.
- `\bgpatternblock{color}` — pattern block headers (`tblock`, `talert`, `texample`, `tpatternbox`), where each block supplies its own color at render time.

# Advanced: Custom Background Pattern (III)

The 12 built-in pattern definitions in `beamerinnerthemewildcat.sty` (lines 31–386) are a useful reference and starting point.

`\bgpatternplain` (line 384) is the simplest — a single `\fill` command — and is a good template to copy and modify. The other patterns show progressively more complex TikZ techniques: coordinate scaling, `\foreach` loops, random seeds, scopes, and drop shadows.

The `\bgpatternblock` variant used in `tblock` headers is typically identical to `\bgpattern`, just registered separately so the theme can pass the block's color through as an argument.

# Advanced: Community Background Patterns

A pattern template and example patterns are available in the `bg-gallery/` folder. To use one, add a single `\input` line to your preamble *after* `\usetheme{wildcat}`:

```
\usetheme{wildcat}
\input{bg-gallery/bg-template}   % swap in any pattern file
```

Use `\input{}`, not `\include{}` — pattern commands must be set in the preamble, and `\include` only works in the document body.

**Want to contribute?** Copy `bg-gallery/bg-template.tex`, replace the drawing commands with your own TikZ, and open a pull request at `github.com/aarondwolf/wildcat`. Community patterns are welcome!